Statics Truss Force Analysis

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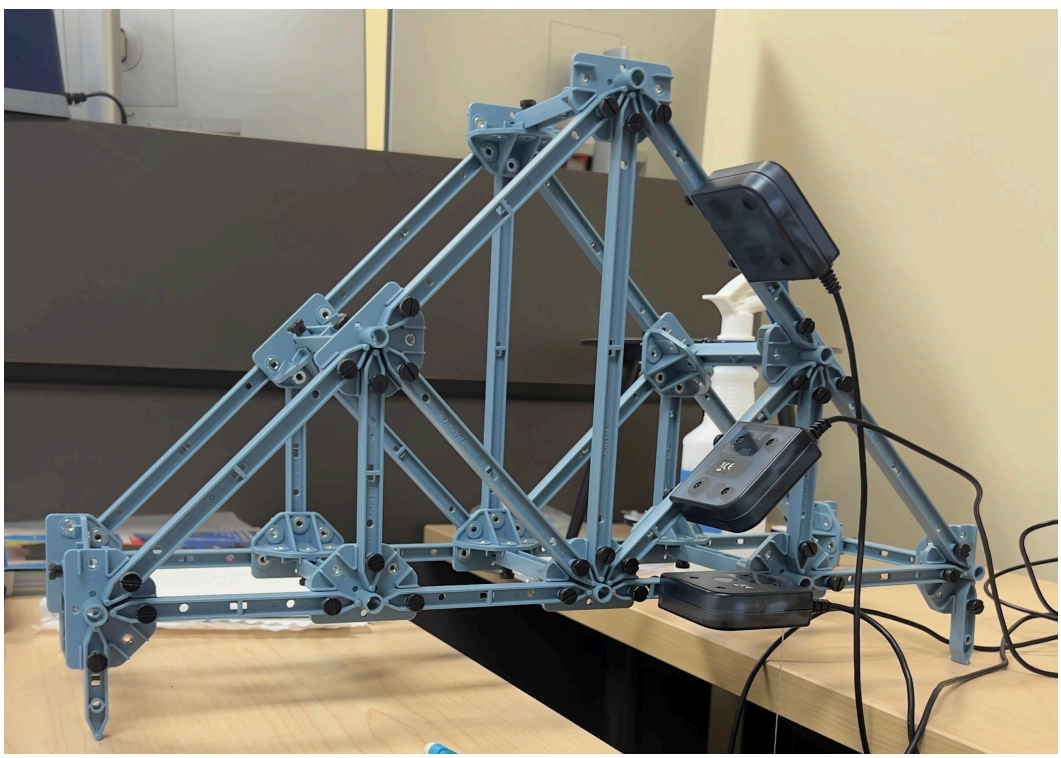
Objective

The goal of the experiment was to perform a truss analysis and compare the theoretical results with the experimental results. The focus was on applying the method of sections to a truss in the real world and comparing the theoretical results that were done on paper using the method of sections and equilibrium equations to the results from the force meters on the three members of the truss.

Theory

In statics, the truss will always be in equilibrium, so each joint will be in equilibrium. The equilibrium equations ($\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M_o = 0$) are used to find the forces in each member, and the truss is cut using the method of sections to find the force in the three desired members. All the forces are assumed to be in tension initially, the members are assumed to be pin-connected at both ends, and there are no axial loads on the members.

Hardware

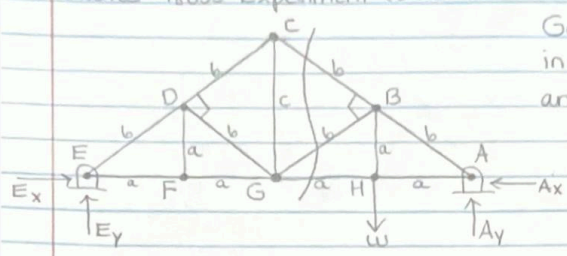


The truss has a total of eight pin connections and thirteen members. The truss is being held by pin connections on the bottom left and bottom right. The force meters are reading the force in the members CB, BG, and HG (from top to bottom), respectively.

Theoretical Work

Statics Truss Experiment (Quiz 7)

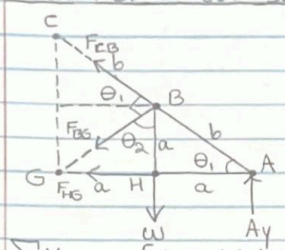
Goal: Find the forces in the members CB, BG, and HG.



$\sum M_E = 0 = [(A_y)(4m) + (-w)(3m)]$, $A_y = 0.75w$

$\sum F_x = 0 = [(A_x + E_x)]$, $A_x = 0$

Draw FBD of cut section



$\theta_1 = \tan^{-1}(\frac{a}{a}) = 45^\circ$

$\theta_2 = \tan^{-1}(\frac{a}{a}) = 45^\circ$

$F_{HG} = 0.75w$

$F_{BG} = -\frac{w}{\sqrt{2}}$

$F_{CB} = -0.3536w$

$\sum M_B = 0 = [(-F_{HG})(a) + (0.75w)(a)]$, $F_{HG} = 0.75w$

$\sum M_A = 0 = [(w)(a) + (F_{BG}\cos 45^\circ)(a) + (F_{BG}\sin 45^\circ)(a)]$

$F_{BG} = -\frac{w}{\sqrt{2}}$

$\sum M_G = 0 = [(0.75w)(2a) + (-w)(a) + (F_{CB}\cos 45^\circ)(a) + (F_{CB}\sin 45^\circ)(a)]$, $F_{CB} = -0.3536w$

Now given $w = 200g = 0.2kg$, $F_w = (0.2kg)(9.81 m/s^2) = 1.962N$

Theoretical Results:

$F_{HG} = 0.75w = 0.75(1.962N) = 1.47N$

$F_{BG} = -\frac{w}{\sqrt{2}} = -\frac{1.962N}{\sqrt{2}} = -1.39N$

$F_{CB} = -0.3536(1.962N) = -0.694N$

Theoretical results are slightly off from the experimental results by around $\pm 0.1N$

Experimental Results:

$F_{HG} = 1.26N$, $F_{BG} = -1.28N$, $F_{CB} = -0.72N$

Final Comparison

Value of Forces from Theoretical Work (F_{CB}, F_{BG}, and F_{HG})	Vale of Forces from Experimental Work (F_{CB}, F_{BG}, and F_{HG})	Percent Error
0.694 N (C)	0.720 N (C)	3.75%
1.39 N (C)	1.28 N (C)	7.91%
1.47 N (T)	1.26 N (T)	14.3%